Recommended Assessment

Encoders

Exploring Decoding Logic

1. When turning the encoder clockwise, do encoder counts go up or down? What about counterclockwise?

2. How many counts are in a full rotation in X1, X2 and X4 decoding? How many tactile bumps were in the encoder? Does this affect the way it is used to read?

Decoding	X1	X2	X4
Counts per revolution			

3. When do the encoder pulses trigger a change in encoder counts? Consider the rising and/or falling edges in both the A and B signals. Describe the encoder counting for each type of decoding.

4. What is the resolution, or minimum angle that can be read, of the encoder on the three different decoding types?

5. How do you convert from counts to degrees and radians? Show the screenshots of your printed outputs from the lab.

6. Does the Quadrature Output Table in the datasheet match what you saw when rotating the encoder? Why?

Real Life Scenarios

7. You are tasked to select an incremental encoder for measuring the position of a DC motor. You require a minimum resolution of 0.01 radians. Furthermore, your data acquisition system (DAQ) and software can only perform X2 decoding. What are the minimum points per revolution (PPR) required to meet this design constraint?

8. As noted in the background section, a counter is used to count the pulses generated by an encoder. Counters are specified using a bit size. For example, a 24-bit counter can hold values up to $2^{24} = 16{,}777{,}216$ counts. When a counter exceeds its limit, it will wrap or overflow.

   Assume that you have a 1,000 PPR encoder (meaning pulses in 1 channel) and a 24-bit counter at your disposal. You plan to use the encoder to monitor the rotational velocity of a shaft that rotates at a maximum speed of 500 rpm. Your DAQ performs X4 decoding. Calculate the minimum duration that the counter can monitor the shaft's speed before it overflows.

9. In this lab, we explored inputs from a quadrature encoder, which reads pulses from 2 output channels that are out of phase. Consider another type of encoder with only a single output channel. What can you measure with a quadrature encoder but not with a single-channel encoder? When would you choose one type over the other?

Encoder Cheat Sheet

What is it?	
What does it measure directly?	
What can you measure indirectly?	
Where are they used?	
Pros	
Cons	
Notes/ Important things to remember	